

P59. Q1

$$\dot{x} = \frac{dx}{dt} = 4x + y - x(x^2 + y^2)$$

$$\dot{y} = \frac{dy}{dt} = 4y - x - y(x^2 + y^2)$$

POLAR COORDINATES

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$

CARTESIAN COORDINATES

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\dot{r} = \frac{d}{dt}(\sqrt{x^2 + y^2}) = \frac{\frac{d}{dt}(x^2 + y^2)}{2\sqrt{x^2 + y^2}} = \frac{x\dot{x} + y\dot{y}}{r}$$

$$\dot{\theta} = \frac{d}{dt}(\arctan(\frac{y}{x})) = \frac{\frac{d}{dt}(\frac{y}{x})}{1 + (\frac{y}{x})^2} = \frac{x\dot{y} - y\dot{x}}{r^2}$$

$$\dot{x} = 4r \cos \theta + r \sin \theta - r^3 \cos \theta$$

$$\dot{y} = 4r \sin \theta - r \cos \theta - r^3 \sin \theta$$

$$\dot{r} = r \cos \theta [4r \cos \theta + r \sin \theta - r^3 \cos \theta] + r \sin \theta [4r \sin \theta - r \cos \theta - r^3 \sin \theta]$$

$$\dot{\theta} = r \cos \theta [4r \sin \theta - r \cos \theta - r^3 \sin \theta] - r \sin \theta [4r \cos \theta + r \sin \theta - r^3 \cos \theta]$$

$$\dot{r} = 4r^2 \cos^2 \theta + r^2 \cos \theta \sin \theta - r^4 \cos^2 \theta + 4r^2 \sin^2 \theta - r^2 \cos \theta \sin \theta - r^4 \sin^2 \theta$$

$$\dot{\theta} = 4r^2 \sin \theta \cos \theta - r^2 \cos^2 \theta - r^4 \sin \theta \cos \theta - 4r^2 \cos \theta \sin \theta - r^2 \sin^2 \theta + r^4 \cos \theta \sin \theta$$

$$\dot{r} = 4r - r^3$$

$$\dot{\theta} = -1$$

orbit clockwise

E.P.

$$4r - r^3 = 0$$

$$r(4 - r^2) = 0$$

$$r(2 - r)(2 + r) = 0$$

$$r = 0 \quad \vee \quad r = 2 \quad \vee \quad r = -2$$

$$x^2 + y^2 = 4 \text{ is a limit cycle}$$

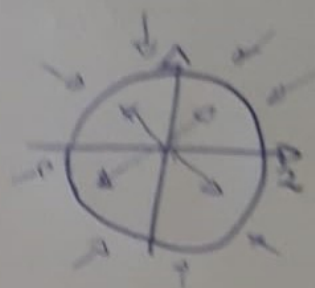
$$0 < r < 2 \quad (I)$$

$$r(2 - r)(2 + r) > 0$$

$$r > 2 \quad (II)$$

$$\text{in } (I) \quad \dot{r} > 0$$

$$\text{in } (II) \quad \dot{r} < 0$$



Q2.

$$\frac{dx}{dt} = 2x + 2y - x(2x^2 + y^2)$$

$$\frac{dy}{dt} = -2x + y - y(x^2 + 2y^2)$$

E.P. (0,0)

JACOBIAN

$$J = \begin{bmatrix} 2 - 6x^2 - y^2 & 2 - 2xy \\ 2 - 2xy & 1 - x^2 - 6y^2 \end{bmatrix}$$

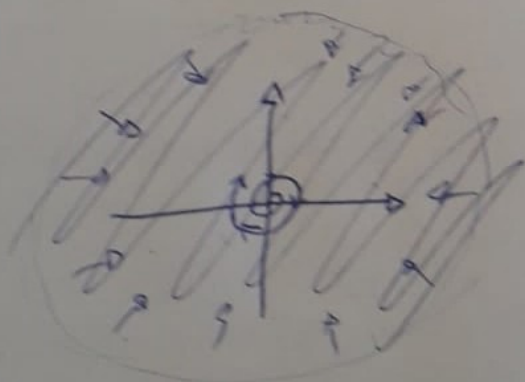
$$J(0,0) = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$$

$$\tau = 3$$

$$\Delta = 2 + 4 = 6$$

$$\tau^2 - 4\Delta = 9 - 24 = -15$$

$\Rightarrow$ unstable spiral



ASYMPTOTIC ANALYSIS

For $|x| \gg 0$ and $|y| \gg 0$

$$\dot{x} = -x(2x^2 + y^2)$$

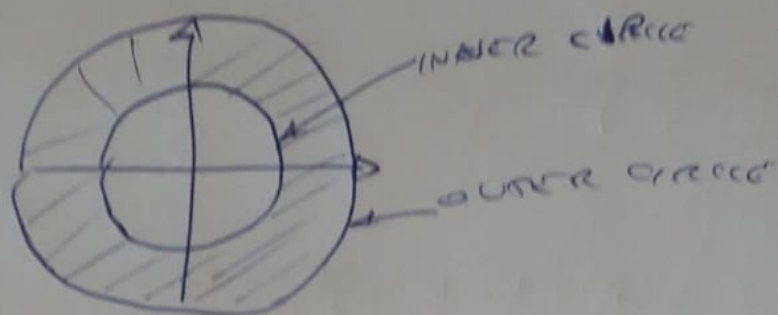
$$\dot{y} = -y(x^2 + 2y^2)$$

$$x > 0, y > 0 \Rightarrow \dot{x} < 0, \dot{y} < 0$$

$$x > 0, y < 0 \Rightarrow \dot{x} < 0, \dot{y} > 0$$

$$x < 0, y > 0 \Rightarrow \dot{x} > 0, \dot{y} < 0$$

$$x < 0, y < 0 \Rightarrow \dot{x} > 0, \dot{y} > 0$$



$$\vec{r} = \frac{x\vec{i} + y\vec{j}}{r}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\vec{i} = 2x + 2y - x(2x^2 + y^2) = 2x + 2y - 2x^3 - xy^2$$

$$\vec{j} = -2x + 2y - y(x^2 + 2y^2) = -2x + 2y - yx^2 - 2y^3$$

$$\vec{r} = \frac{2x^2 + 2xy - 2x^4 - x^2y^2 - 2xy + y^2 - 2y^2x^2 - 2y^4}{r}$$

$$\vec{r} = \frac{2r^2 \cos^2 \theta - 2r^4 \cos^4 \theta + r^2 \sin^2 \theta - 2r^4 \sin^2 \theta \cos^2 \theta - 2r^4 \sin^4 \theta}{r}$$

$$\vec{r} = \frac{2r^2 \cos^2 \theta + r^2 \sin^2 \theta - 2r^4 \sin^2 \theta \cos^2 \theta - 2r^4 \cos^4 \theta - 2r^4 \sin^4 \theta}{r}$$

$$\vec{r} = \frac{2r^2 \cos^2 \theta + r^2 \sin^2 \theta - 2r^4 \sin^2 \theta \cos^2 \theta - 2r^4 (\cos^4 \theta + (1 - \cos^2 \theta)^2)}{r}$$

$$\vec{r} = \frac{2r^2 \cos^2 \theta + r^2 + r^2 \cos^2 \theta + 2r^4 \cos^2 \theta + 2r^4 \cos^4 \theta - 2r^4 \cos^4 \theta - 2r^4 + 4r^4 \cos^2 \theta - 2r^4 \cos^4 \theta}{r}$$

$$\vec{r} = \frac{r^2 (1 + \cos^2 \theta) - 2r^4 (1 - \cos^2 \theta + \cos^4 \theta)}{r}$$

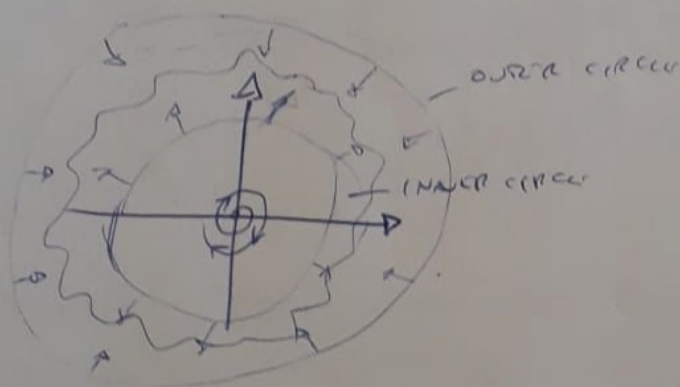
$$\dot{r}^2 = r^4 [(1 + \cos^2 \theta) - 2r^2 (1 - \cos^2 \theta + \cos^4 \theta)]$$

$$r > 0 \quad (1 + \cos^2 \theta) - 2r^2 (1 - \cos^2 \theta + \cos^4 \theta) > 0$$

$$\begin{array}{l} q(\theta) \\ p(\theta) \end{array} \quad q(\theta) - 2r^2 p(\theta) > 0$$

if $q(\theta) > 2r^2 p(\theta)$ r is increasing

if $q(\theta) < 2r^2 p(\theta)$ r is decreasing



$$\bullet \quad q(\theta) = 1 + \cos^2 \theta \quad q_{\min} = 1$$

$$q_{\max} = 2$$

$$\bullet \quad p(\theta) = 1 - \cos^2 \theta + \cos^4 \theta$$

$$p'(\theta) = 2 \cos \theta \sin \theta - 4 \cos^3 \theta \sin \theta = 2 \cos \theta \sin \theta (1 - 2 \cos^2 \theta)$$

$$2 \cos \theta \sin \theta (1 - 2 \cos^2 \theta) = 0$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = 0 \quad \vee \quad \sin \theta = 0 \quad \vee \quad 1 - 2 \cos^2 \theta = 0$$

$$p = 1$$

$$p = 1$$

$$p = 1 - \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$q(0) > 2r^2 p(0)$$

$$q_{\min} = 1$$

$$q_{\max} = 2$$

$$p_{\min} = \frac{3}{4}$$

$$p_{\max} = 1$$

LARGEST RADIUS WE CAN GET FOR THE INNER CIRCLE

$$r_i > 0$$

$$q_{\min} > 2r_i^2 p_{\max}$$

$$1 > 2r_i^2 \cdot 1$$

$$\boxed{r_i < \frac{1}{\sqrt{2}}}$$

SMALLEST RADIUS WE CAN GET FOR THE OUTER CIRCLE

$$r_o < 0$$

$$q_{\max} < 2r_o^2 p_{\min}$$

$$2 < 2r_o^2 \cdot \frac{3}{4}$$

$$\boxed{r_o > \frac{2}{\sqrt{3}}}$$

ANNULAR REGION

$$\boxed{\frac{1}{\sqrt{2}} < x^2 + y^2 < \frac{2}{\sqrt{3}}}$$